GSE 518
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**Suggested Solutions to Problem Set 6**

**1.**
**(a)** We know that we can derive the joint density $f(x,y)$ from the joint cdf $F(x,y)$ by using: $f(x,y) = \frac{\partial^2 F(x,y)}{\partial x \partial y}$. In this case we have $\partial F(x,y)/\partial x = (1-e^{-y})e^{-x}$ and $\partial^2 F(x,y)/\partial x \partial y = e^{-x}e^{-y}$. Hence, $f(x,y) = e^{-x}e^{-y}$ for $x > 0$, $y > 0$ and zero otherwise.
**(b)** In this case we can write the joint density of $f(x,y)$ as the product of a function of $x$ (say $g(x) = e^{-x}$) times a function of $y$ (say $h(y) = e^{-y}$). Therefore, by the Factorization Theorem (Lemma 4.2.7) we have that $X$ and $Y$ are independent.

**2.**
**(a)** We can calculate this probability using the joint pdf as:
$\Pr(X < 0.25, Y < 1) = \int_0^1 \int_0^{0.25} f(x,y)\,dxdy = \int_0^1 \int_0^{1/4} 0.25(2x+y)\,dxdy$
$= 0.25\int_0^1 \left[(x^2+xy)\big|_0^{1/4}\right] dy = 0.25\int_0^1 \left(\frac{1}{16}+\frac{1}{4}y\right) dy = \frac{1}{4}\left[\left(\frac{1}{16}y+\frac{1}{8}y^2\right)\Big|_0^1\right] = \frac{1}{4}\left[\frac{1}{16}+\frac{1}{8}\right] = \frac{3}{64} = 0.047$
**(b)** To find the marginal of $X$ we integrate the joint pdf over $Y$:
$f_X(x) = \int_0^2 f(x,y)\,dy = \int_0^2 0.25(2x+y)\,dy = \frac{1}{4}\left[2xy+\frac{1}{2}y^2\right]\Big|_0^2 = \frac{1}{4}[4x+2] = x+\frac{1}{2}$. Therefore, $f_X(x) = x+\frac{1}{2}$ if $0 < x < 1$ and zero otherwise.
To find the marginal of $Y$ we integrate the joint pdf over $X$:
$f_Y(y) = \int_0^1 f(x,y)\,dx = \int_0^1 0.25(2x+y)\,dx = \frac{1}{4}[x^2+xy]\big|_0^1 = \frac{1}{4}[1+y]$. Therefore, $f_Y(y) = \frac{1}{4}[1+y]$ if $0 < y < 2$ and zero otherwise.
**(c)** $E(X) = \int_0^1 x f_X(x)\,dx = \int_0^1 x\left(x+\frac{1}{2}\right)dx = \int_0^1 \left(x^2+\frac{x}{2}\right)dx = \left(\frac{x^3}{3}+\frac{x^2}{4}\right)\Big|_0^1 = \frac{1}{3}+\frac{1}{4} = \frac{7}{12}$.
**(d)** Note that in this case we cannot write the joint density $f(x,y) = 0.5x + 0.25y$ as the product of two functions $g(x)$ and $h(y)$. Therefore, by the Factorization Theorem (Lemma 4.2.7) we have that $X$ and $Y$ are not independent.
**(e)** By definition we have that $f_{Y|X}(y|X=x) = f(x,y)/f_X(x)$. Using our result from part (b) we have that $f_{Y|X}(y|X=x) = f(x,y)/f_X(x) = 0.25(2x+y)/(x+0.5)$, for $0 < y < 2$ and zero otherwise.
**(f)** As in part (e), we have that by definition $f_{X|Y}(x|Y=y) = f(x,y)/f_Y(y)$. Using our result from part (b) we have that $f_{X|Y}(x|Y=y) = f(x,y)/f_Y(y) = 0.25(2x+y)/0.25(1+y) = (2x+y)/(1+y)$, for $0 < x < 1$ and zero otherwise.
**(g)** Using our result from part (f) we have that the conditional pdf of $X$ given $Y = 1$ is given by: $f_{X|Y}(x|Y=1) = \frac{1}{2}(2x+1) = x+\frac{1}{2}$. Hence, the required probability is given by
$\Pr(X < 0.25|Y=1) = \int_0^{0.25} f_{X|Y}(x|Y=1)\,dx = \int_0^{0.25}\left(x+\frac{1}{2}\right)dx == \left[\frac{x^2}{2}+\frac{x}{2}\right]_0^{1/4} = \frac{1}{32}+\frac{1}{8} = \frac{5}{32} = 0.15625$
**(h)** Using the conditional pdf $f_{X|Y}(x|Y=y) = (2x+y)/(1+y)$ from part (f) we find (note that here $Y = y$ is given and the only random variable is $X$)

$E(X|Y=y) = \int_0^1 x f_{X|Y}(x|Y=y)\,dx = \int_0^1 x(2x+y)/(1+y)\,dx = \frac{1}{1+y}\int_0^1 (2x^2+xy)\,dx = \frac{1}{1+y}\left(\frac{2}{3}x^3+\frac{1}{2}x^2y\right)\Big|_0^1 = \frac{1}{1+y}\left(\frac{2}{3}+\frac{1}{2}y\right)$. Hence, $E(X|Y=y) = \frac{1}{1+y}\left(\frac{2}{3}+\frac{1}{2}y\right)$.
**(i)** Using the marginal density of $Y$ from part (b), we would calculate $E[E(X|Y=y)]$ as $E[E(X|Y=y)] = \int_0^2 E(X|Y=y) f_Y(y)\,dy = \int_0^2 \frac{1}{1+y}\left(\frac{2}{3}+\frac{1}{2}y\right)\left(\frac{1}{4}[1+y]\right)dy$. It is *very important* to note that $E(X|Y=y)$ is only a function of $Y$ ($X$ was already integrated out in part (h)); therefore, the outer expectation is taken with respect to the marginal density of $Y$.
**(j)** From the Law of Iterated Expectations we know that $E(X) = E[E(X|Y=y)]$. Thus, from our result in (c) we know $E(X) = E[E(X|Y=y)] = \frac{7}{12}$.

**3.**
**(a)** Using the definition of covariance we have:
$Cov(aX,Y) = E\{[aX - E(aX)][Y - E(Y)]\} = E\{[aX - aE(X)][Y - E(Y)]\} =$
$E\{a[X - E(X)][Y - E(Y)]\} = aE\{[X - E(X)][Y - E(Y)]\} = aCov(X,Y)$
**(b)** By definition, and following similar steps as the ones in part (a), we have:
$Cov(aW+bX, cY+dZ) = E\{[aW+bX - E(aW+bX)][cY+dZ - E(cY+dZ)]\}$
$= E\{[a(W-EW)+b(X-EX)][c(Y-EY)+d(Z-EZ)]\}$
$= E\{[ac(W-EW)(Y-EY)+ad(W-EW)(Z-EZ)+bc(X-EX)(Y-EY)]$
$+bd(X-EX)(Z-EZ)\}$
$= acCov(W,Y) + adCov(W,Z) + bcCov(X,Y) + bdCov(X,Z)$

**4.** By definition we know that the correlation between $U$ and $V$ is given by $\rho_{UV} = Cov(U,V)/\sigma_u\sigma_v$. We now find an expression for each of the terms in this formula using the fact that $X$ and $Y$ follow a bivariate normal distribution with parameters $N\left(\mu_x,\mu_y,\sigma_x^2,\sigma_y^2,\rho_{xy}\right)$. Using the result in 3(b) we know that $Cov(U,V) = Cov(X+Y,X-Y) = Cov(X,X) - Cov(Y,Y) = Var(X) - Var(Y) = \sigma_x^2 - \sigma_y^2$. We also have $Var(U) = \sigma_u^2 = Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y) = \sigma_x^2 + \sigma_y^2 + 2\rho_{xy}\sigma_x\sigma_y$, where we used the fact that $Cov(X,Y) = \rho_{xy}\sigma_x\sigma_y$. Following similar steps we find $Var(V) = \sigma_v^2 = Var(X-Y) = \sigma_x^2 + \sigma_y^2 - 2\rho_{xy}\sigma_x\sigma_y$. Hence, we can write $\rho_{UV}$ as
$\rho_{UV} = \frac{Cov(U,V)}{\sigma_u\sigma_v} = \frac{\sigma_x^2-\sigma_y^2}{\left[\left(\sigma_x^2+\sigma_y^2+2\rho_{xy}\sigma_x\sigma_y\right)\left(\sigma_x^2+\sigma_y^2-2\rho_{xy}\sigma_x\sigma_y\right)\right]^{1/2}} = \frac{\sigma_x^2-\sigma_y^2}{\left[\left(\sigma_x^2+\sigma_y^2\right)^2-4\rho_{xy}^2\sigma_x^2\sigma_y^2\right]^{1/2}}$

**5.** Each $X_i$ follows a $Chi\text{-}Squared(p_i)$ distribution, and therefore, its mgf is given by $M_{X_i}(t) = \left(\frac{1}{1-2t}\right)^{p_i/2}$. From the implications of independence seen in class, we know that since $X_1, X_2, \ldots, X_n$ are mutually independent then the mgf of $Y = \sum_{i=1}^n X_i$ is given by
$M_Y(t) = \prod_{i=1}^n M_{X_i}(t) = \prod_{i=1}^n \left(\frac{1}{1-2t}\right)^{p_i/2} = \left(\frac{1}{1-2t}\right)^{\sum_{i=1}^n p_i/2}$
The mgf of $Y$ can be recognized as the mgf of a $Chi\text{-}Squared\left(\sum_{i=1}^n p_i\right)$. Therefore, by Theorem 2.3.11 we know that $Y \sim Chi\text{-}Squared\left(\sum_{i=1}^n p_i\right)$. Hence, the sum of independent Chi-squared random variables follows a Chi-squared distribution with the degrees of freedom

equal to the sum of the degrees of freedom.

**6.**

**(a)** First we show that independence implies $\rho = Cov(X,Y)/\sigma_x\sigma_y = 0$. If $X$ and $Y$ are independent we have that $Cov(X,Y) = E(XY) - E(X)E(Y) = E(X)E(Y) - E(X)E(Y) = 0$, which implies $\rho = 0$. Note that we used the fact that $X$ and $Y$ are independent in the second equality.

Now we show that $\rho = 0$ implies independence. By setting $\rho = 0$ we can write the joint pdf $f(x,y)$ as

$$f(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}\exp\left\{-\frac{1}{2}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]\right\}$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}\sigma_x}\exp\left\{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_x}\right)^2\right\}}_{g(x)}\underbrace{\frac{1}{\sqrt{2\pi}\sigma_y}\exp\left\{-\frac{1}{2}\left(\frac{y-\mu_y}{\sigma_y}\right)^2\right\}}_{h(y)} = g(x)\,h(y)$$

Therefore, since we can write the joint density of $f(x,y)$ as the product of a function of $x$ times a function of $y$ we know by the Factorization Theorem (Lemma 4.2.7) that $X$ and $Y$ are independent. In fact, note that in this case we have that $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N\left(\mu_y, \sigma_y^2\right)$.

**(b)** From the question we can take as given that linear combinations of normally distributed random variables are normal. Hence, we only need to calculate the mean and variance of $aX + bY$. The mean is $E(aX+bY) = aE(X) + bE(Y) = a\mu_x + b\mu_y$. And the variance is $Var(aX+bY) = a^2\sigma_x^2 + b^2\sigma_y^2 + 2abCov(X,Y) = a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_x\sigma_y\rho$. Therefore, $aX + bY \sim N\left(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\rho\sigma_x\sigma_y\right)$.